	$1+\tan^2\theta=\sec^2\theta$	From (2), $MP^2+OM^2=OP^2$ Dividing each term on both sides of (2) by OM^2, ($\because OM\neq 0$) we get, $\frac{MP^2}{OM^2}+\frac{OM^2}{OM^2}=\frac{OP^2}{OM^2}$ $\Rightarrow \left(\frac{MP}{OM}\right)^2+\left(\frac{OM}{OM}\right)^2=\left(\frac{OP}{OM}\right)^2$ From (3), $(\tan\theta)^2+1^2=(\sec\theta)^2$ Hence $1+\tan^2\theta=\sec^2\theta$.
	$1+\cot^2\theta=\operatorname{cosec}^2\theta$	In the right angled ΔOMP, we have $\frac{OM}{MP}=\cot\theta,\quad \frac{OP}{MP}=\operatorname{cosec}\theta$ ...(4) From (2), $MP^2+OM^2=OP^2$ Dividing each term on both sides of (2) by MP^2, ($\because MP\neq 0$) we get, $\frac{MP^2}{MP^2}+\frac{OM^2}{MP^2}=\frac{OP^2}{MP^2}$ $\Rightarrow \left(\frac{MP}{MP}\right)^2+\left(\frac{OM}{MP}\right)^2=\left(\frac{OP}{MP}\right)^2$ From (4), $1^2+(\cot\theta)^2=(\operatorname{cosec}\theta)^2$ Hence, $1+\cot^2\theta=\operatorname{cosec}^2\theta$

These identities can also be rewritten as follows.

Identity	Equal forms
$\sin^2\theta+\cos^2\theta=1$	$\sin^2\theta=1-\cos^2\theta$ (or) $\cos^2\theta=1-\sin^2\theta$
$1+\tan^2\theta=\sec^2\theta$	$\tan^2\theta=\sec^2\theta-1$ (or) $\sec^2\theta-\tan^2\theta=1$
$1+\cot^2\theta=\operatorname{cosec}^2\theta$	$\cot^2\theta=\operatorname{cosec}^2\theta-1$ (or) $\operatorname{cosec}^2\theta-\cot^2\theta=1$

Note

Though the above identities are true for any angle θ, we will consider the six trigonometric ratios only for $0^\circ<\theta<90^\circ$

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